EMCal calibration update: data/sim comparisons

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Non-linearity: fiber atenuation

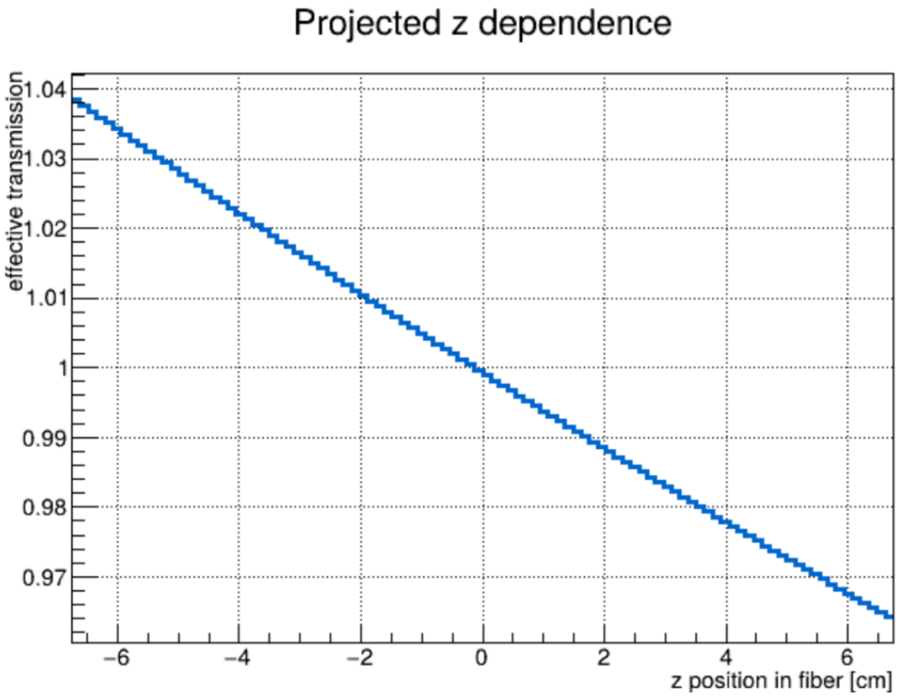
1. **Longitudinal light attenuation in fibers.** Higher-energy showers develop deeper in the calorimeter, increasing the average light path to the SiPM and enhancing attenuation. A compact estimate follows from (i) electromagnetic shower depth scaling $t_{\max} \propto \ln E$, so $\Delta x \sim X_0 \ln E$, and (ii) exponential attenuation $S \propto e^{-\Delta x/\Lambda}$, which gives

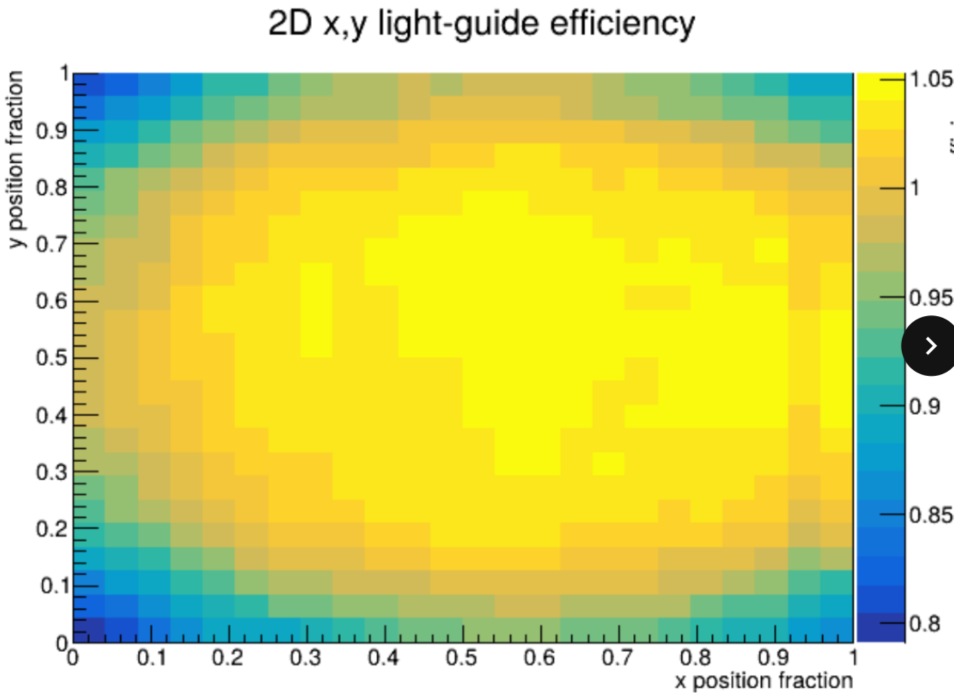
$$S(E) \propto e^{-X_0 \ln(E)/\Lambda} \propto E^{-X_0/\Lambda}, \qquad \frac{S(E)}{E} \propto E^{-X_0/\Lambda}.$$

(Equivalently, the correction factor is the inverse power.) Using an attenuation-length scale of order $\Lambda 1.6$ m (as quoted in the TDR) and normalizing at 1 GeV, a simple depth-scaling estimate gives a response change of approximately 1.0%, 1.3%, and 1.6% at 10, 20, and 40 GeV, respectively.

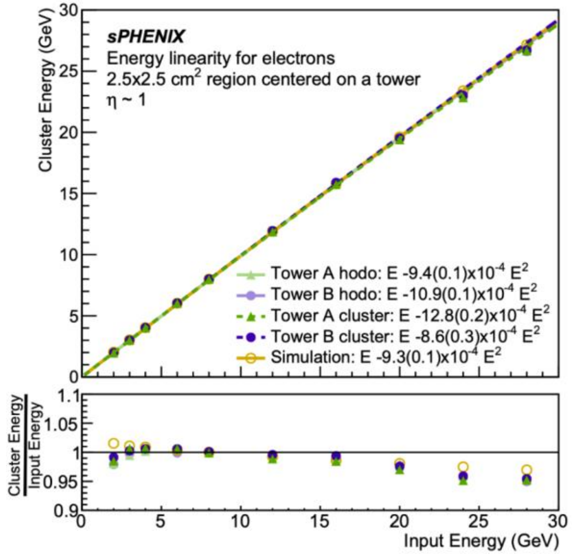
This is implemented in the current MC in the form of a z-dependent light efficiency

Not super important for extrapolating non-linearity out to 30 GeV (where there are no neutral mesons





Non-linearity: pixel occupancy

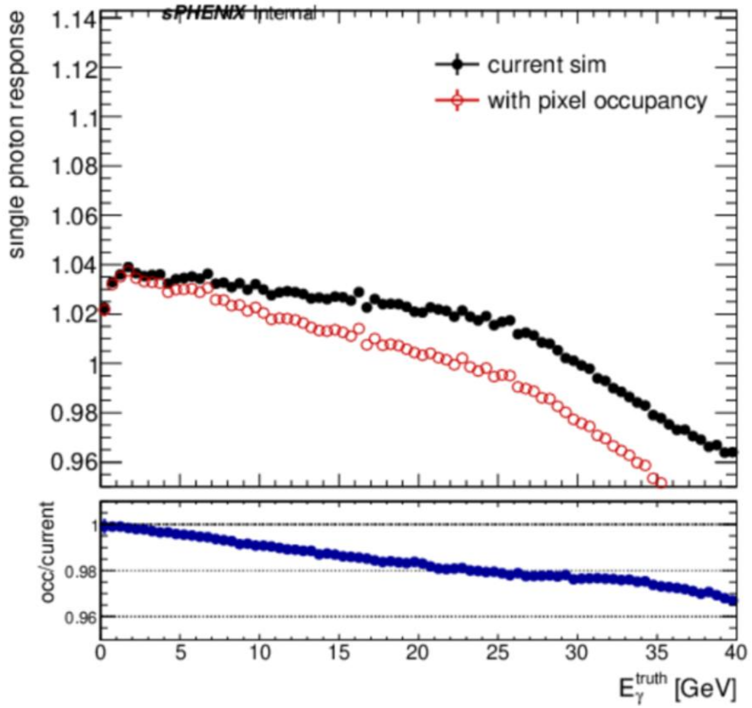


2. **SiPM pixel saturation/occupancy.** As deposited energy increases, the number of fired pixels rises and the probability of multiple photons competing for the same pixel increases. With $N_{\text{pix}} \approx 1.6 \times 10^5$ total pixels (4 SiPMs × 40k each) and $\sim$ 400 photoelectrons per GeV, a first-order occupancy estimate is

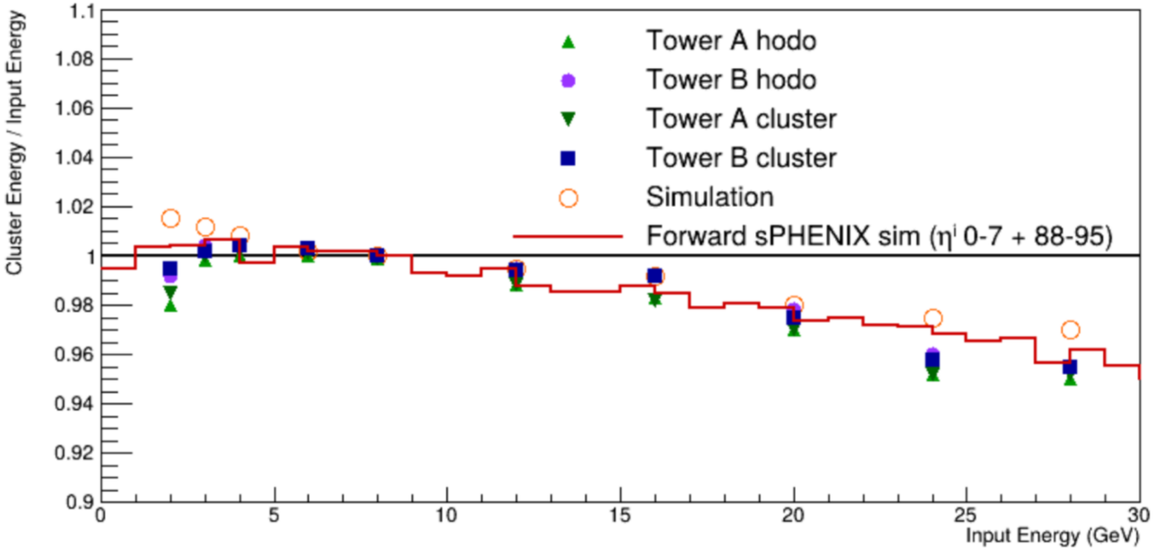
$$\frac{400\,E}{160000 \times 2} \approx 0.125\% \times E,$$

corresponding to roughly 1.25%, 2.5%, and 5% at 10, 20, and 40 GeV.

- **Not in the current sPHENIX MC**
  - Was included in teat beam sim
- **Implemented**
- **Two parameters**
  - Number of electrons per a GeV - 500
  - Number of effective pixels (al pixels)

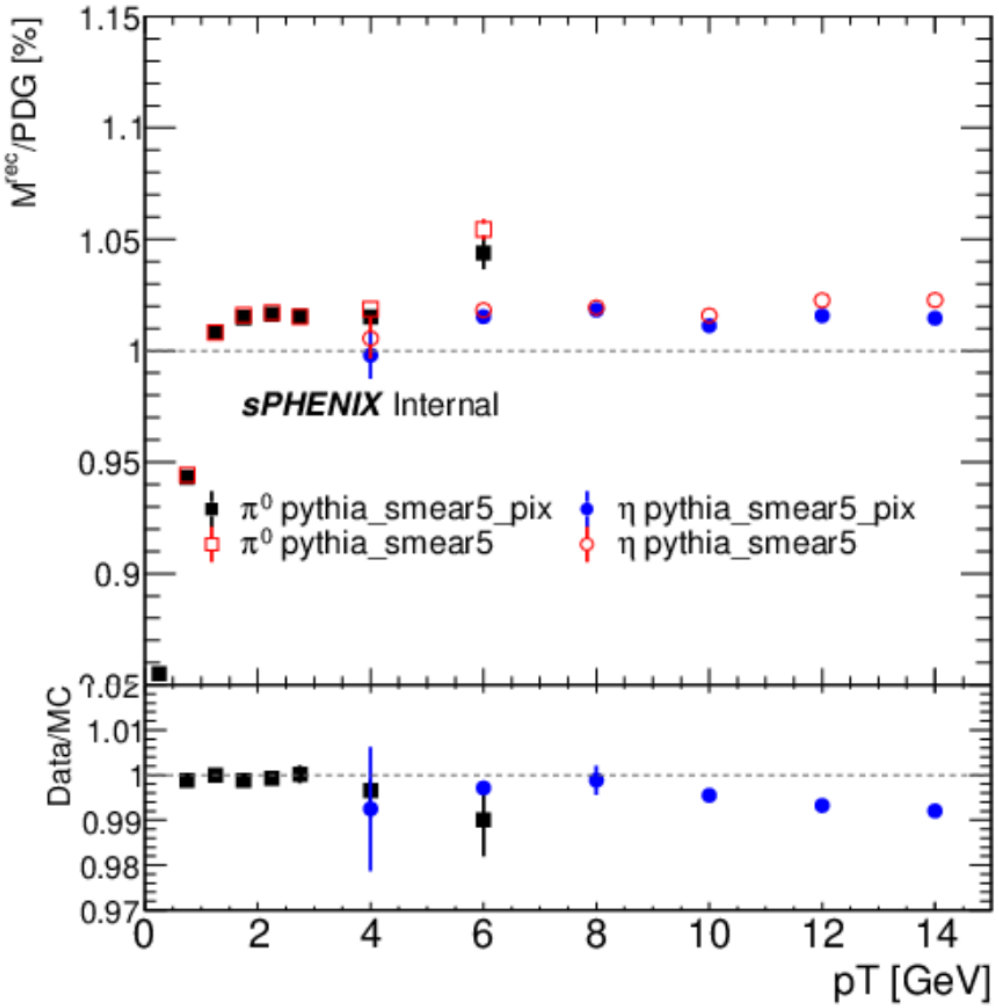


Comparison to test beam results



Effect of pixel occupancy

- ➢Effect on neutral meson mass
- ➢Pythia_smear5 = nominal sim, pythia_smear5-pix = sim with pixel occupancy effect



Selections

- **Data: ana509, GRL**
- **Pythia run 28: main samples MB pythia, Eta3 and Eta8**
- **MBD vertex |z| < 40 cm**
- **Cluster selections**
  - pT1,2 > 0.25 GeV
  - Asym < 0.6

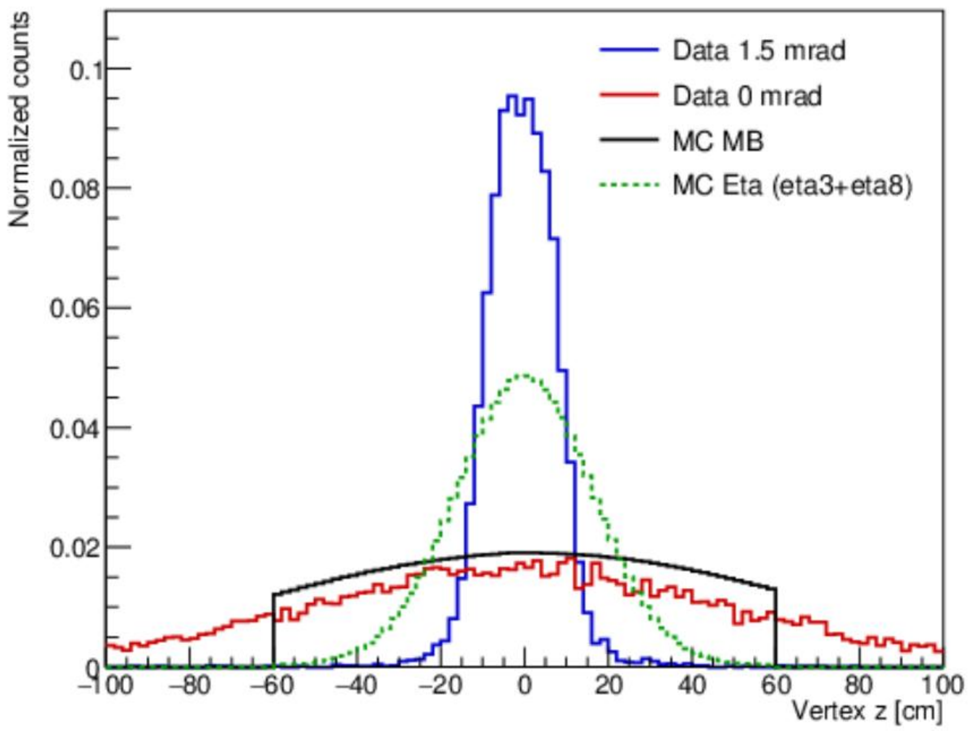
Pileup info

- trigger_index: 12
- weighted_mu: 0.1094917194
- total_scaled_events: 13124488942
- weighted_mu_0mrad (run < 51274): 0.2212110310
- total_scaled_events_0mrad: 3578930503 (runs used: 652)
- weighted_mu_1p5mrad (run >= 51274): 0.0676046307
- total_scaled_events_1p5mrad: 9545558439 (runs used: 812)

- trigger_index: 30
- total_scaled_events: 3188620079
- weighted_mu_0mrad (run < 51274): 0.2441211417
- total_scaled_events_0mrad: 2463468095 (runs used: 725)
- weighted_mu_1p5mrad (run >= 51274): 0.0712799910
- total_scaled_events_1p5mrad: 725151984 (runs used: 680)

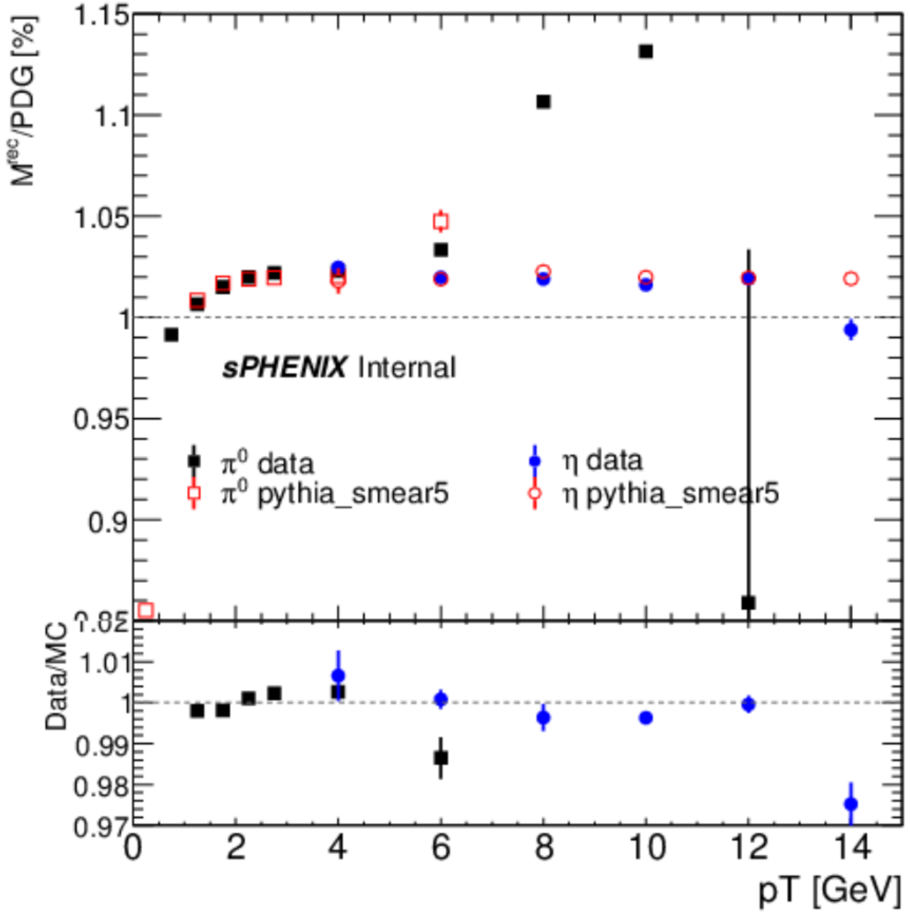
Vertex reweighting and pileup

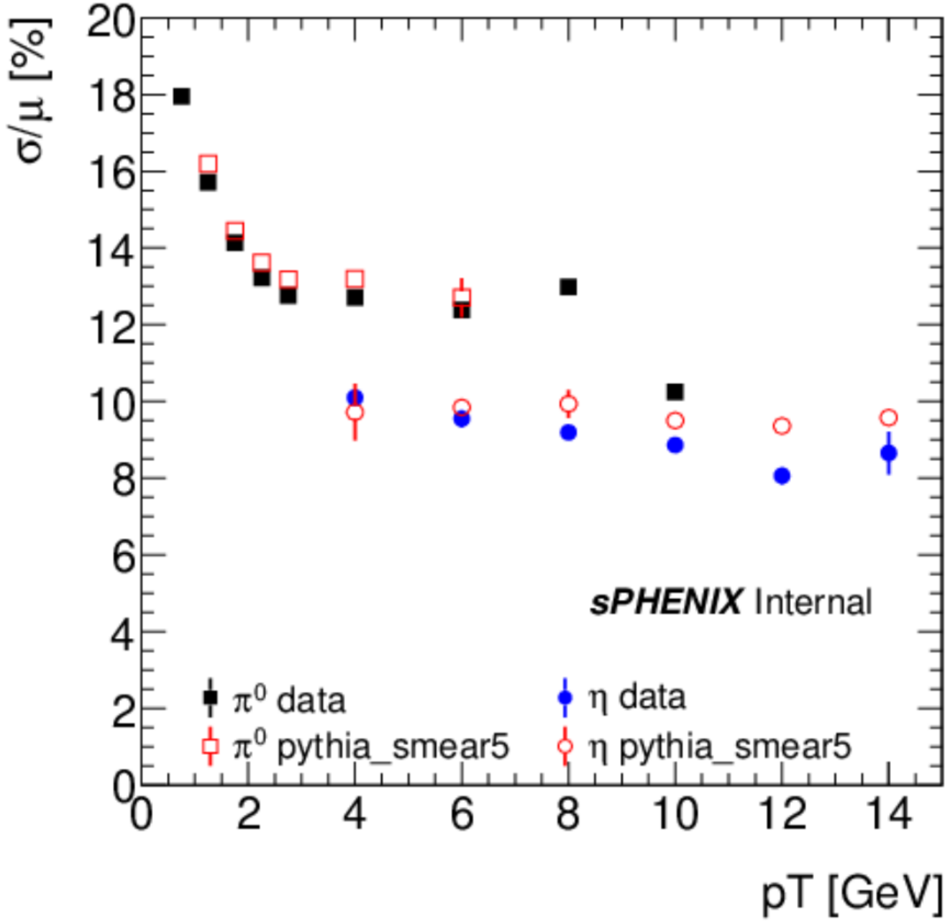
- **MC (MB pythia and EtaX) are reweighted to match 0 and 1.5 mrad**
- **Effects from pileup are added by generating a random synthetic pileup vertex from a gaussian with sigma 50 (15) cm for 0 (1.5) mrad data and this pileup is averaged with the reco vertex**



Data/mc meson comparison 0 mrad

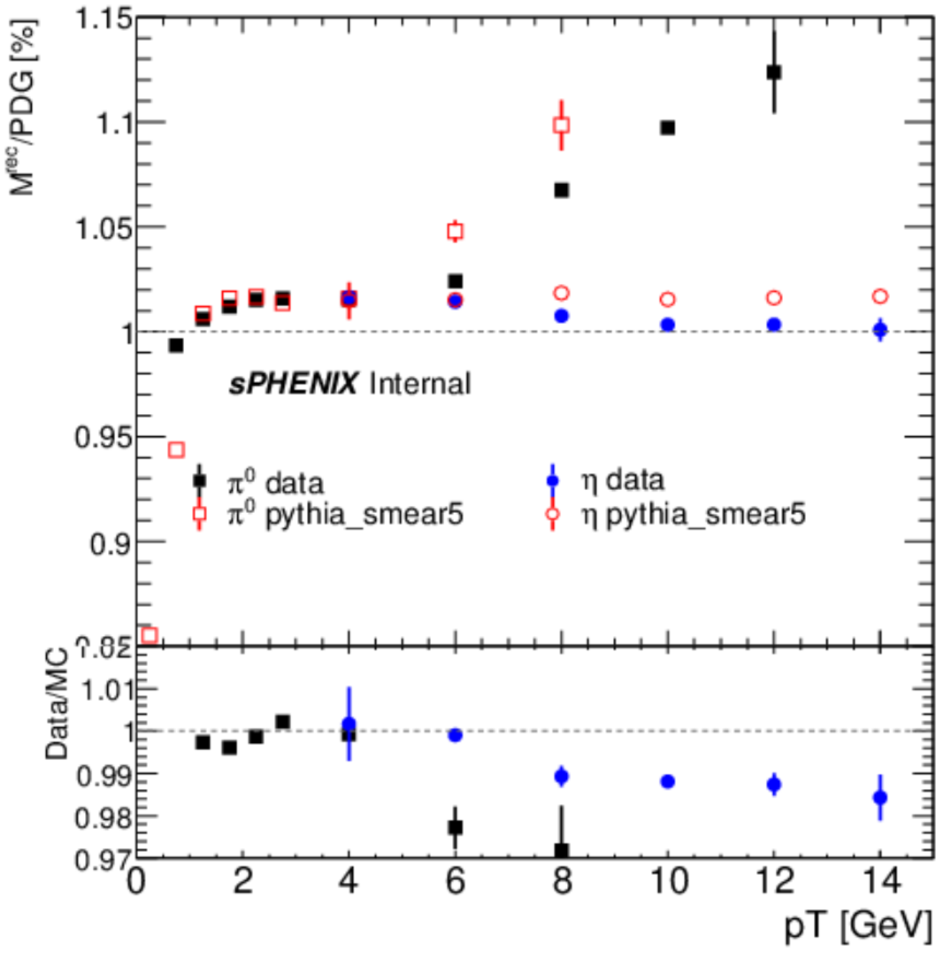
- Good agreement except for last point which is a fitting issue
- Small discrepancy in width, I think this would not lead to large differnce in mass if rectified but am investigating

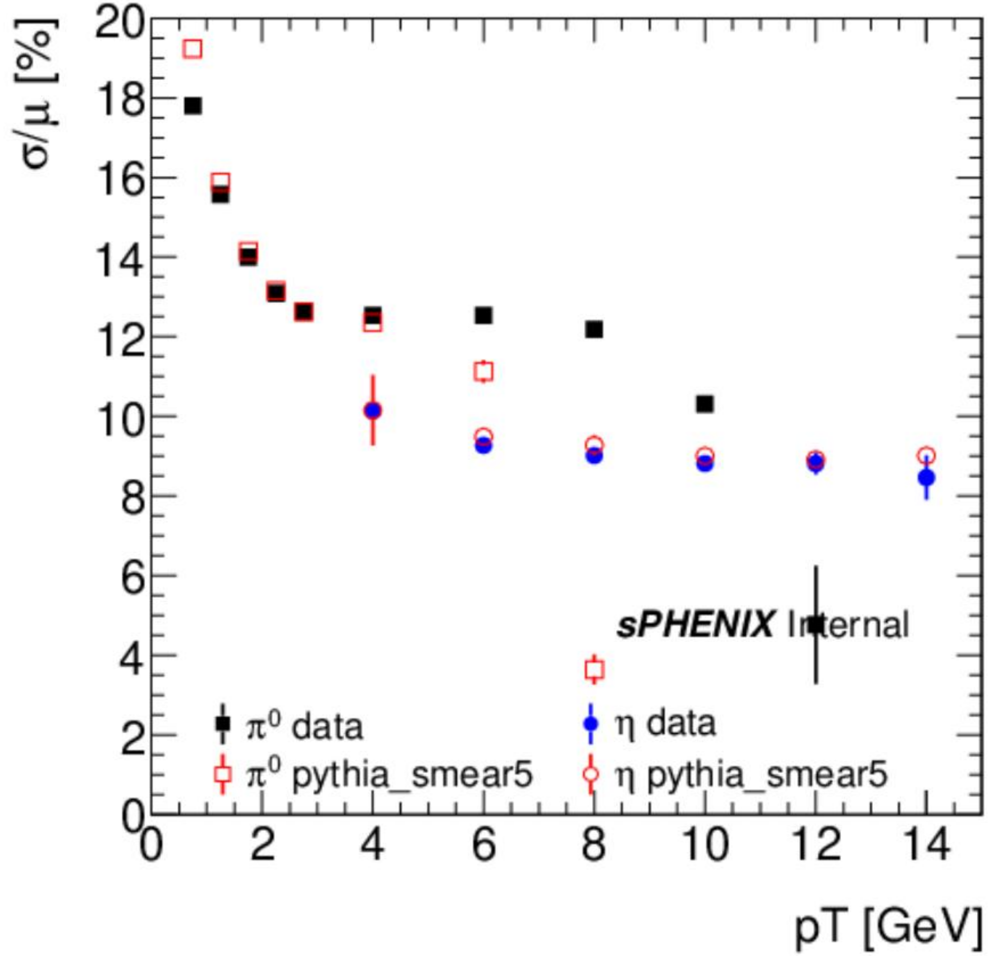




Data/mc meson comparison 1.5 mrad

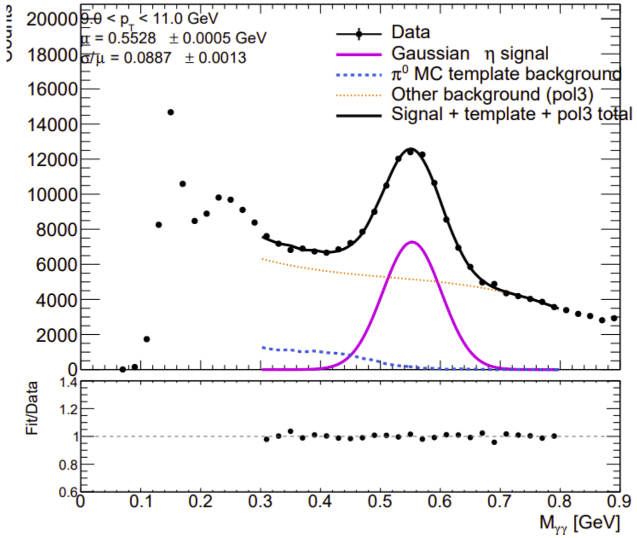
- ➢Non-linearity looks OK, but sudden drop in triggered sample mass
- ➢Small discrepancy in width, I think this would not lead to large differnce in mass if rectified but am investigating

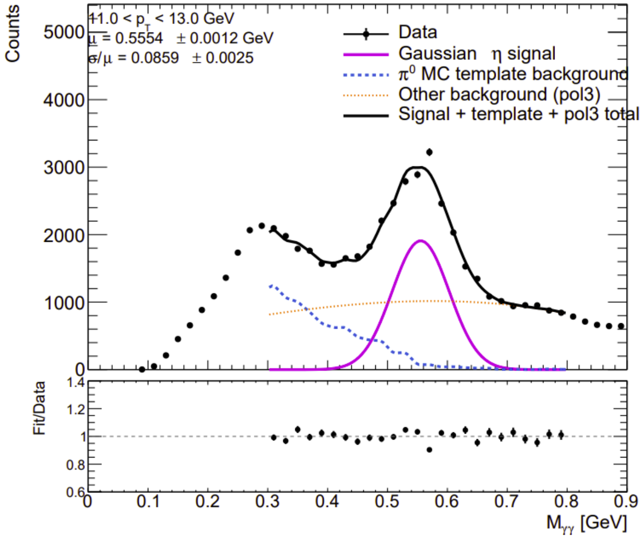


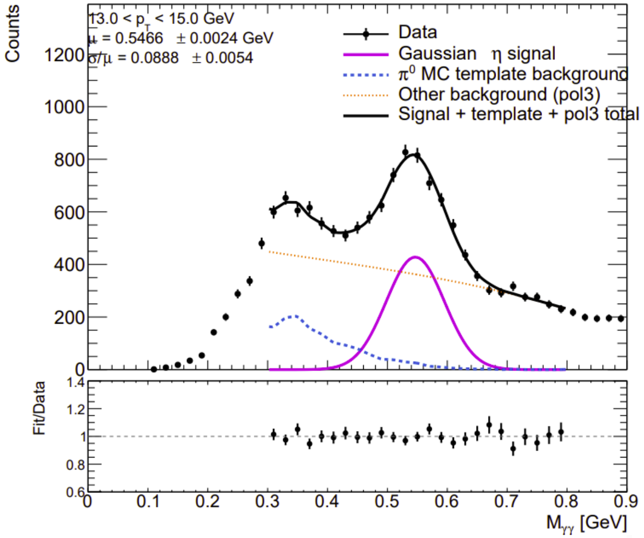


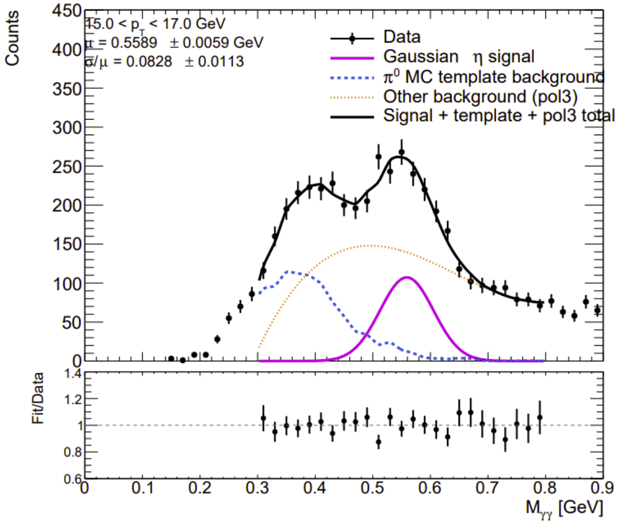
Pi0 template based eta meson fitting

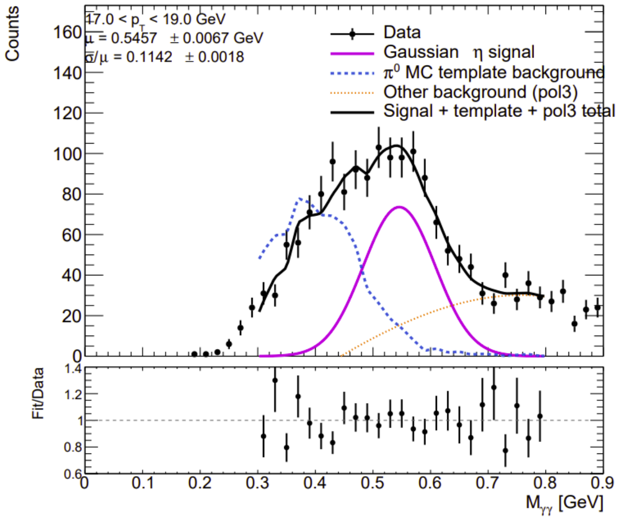
- In an effort to get the furthest pT reach we want to improve the fitting
- Fit with signal gaussian + pi0 template (pi0 gun) + residual background
- This would largely be unnecessary if conversion photon could be rejected with tracking

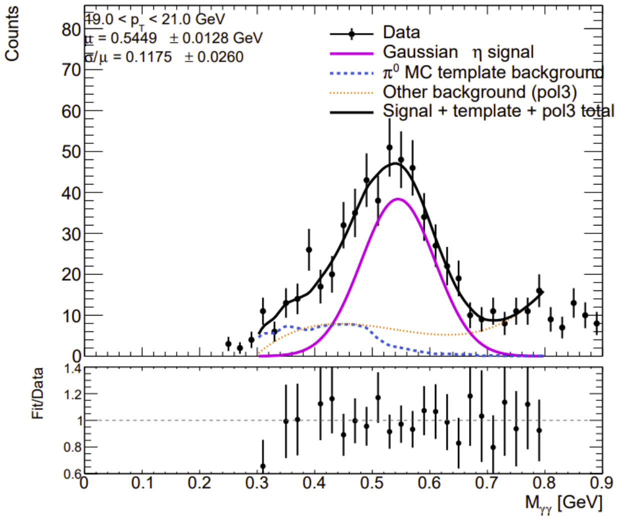




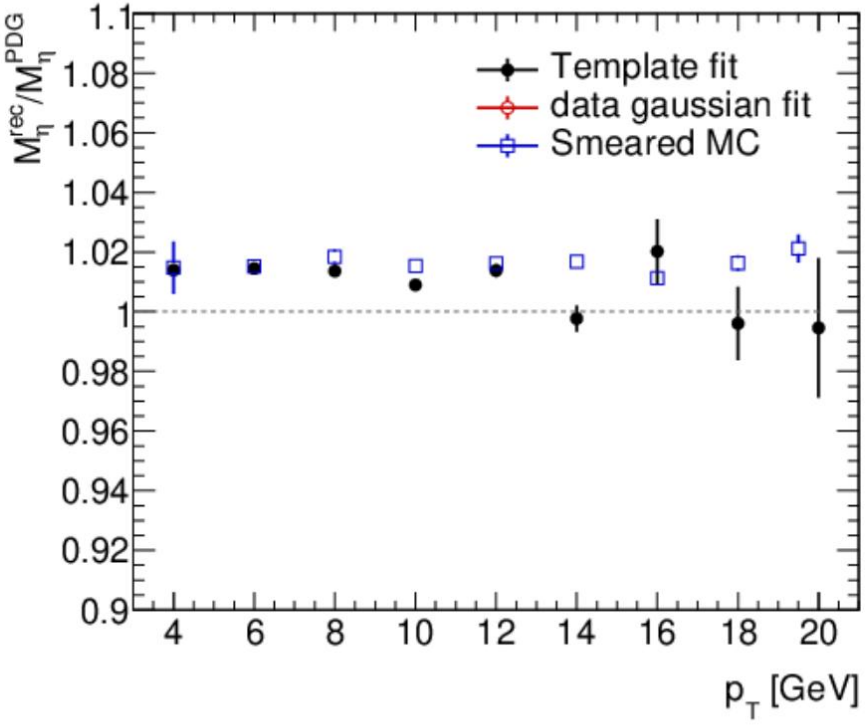








Mass and width results with template fits



Conclusion

- There are still issues with the eta mass but they don’t show some growing discrepancy with pT
- Integrating most of the stats a constraint 2% uncertainty at 20 GeV
- Need to make a discission about PPG12

Omrad vs. 1.5 mrad data

- Omrad (1.5mrad) is estimated to have ~22% (7%) double interactions, which degrades vertex resolution
- This degradation is worse in 0mrad because the vertex width is !50 cm where it is ~15cm in 1.5mrad
- <u>For the rest of this presentation use 1.5 mrad for data/MC comparisons</u>